

# Ellen Poulsen

 Ellen Poulsen |  ellenpoulsen5@gmail.com |  +1 (541) 654-2129

## EDUCATION

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2025 — 2029 **B.S. Aerospace Engineering** GPA: 3.66/4.0  
California Polytechnic State University, San Luis Obispo

2023 Recipient of the Oregon State Seal of Biliteracy in Spanish

**Relevant Coursework:** Introduction to Design and Manufacturing, Welding, Materials Science, Statics & Mechanics, Stars & Planetary Systems, Linear Analysis (expected Spring 2027), Dynamics (expected Spring 2027)

## WORK EXPERIENCE

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**Test Engineering Intern — Empirical Systems Aerospace** *June 2026 — Aug. 2026*  
*San Luis Obispo, CA*

- Developed servo testing procedures and software to analyze frequency response for integration on flight vehicles.
- Programmed servo control mechanisms utilizing both PWM and CAN bus communication protocols.
- Calibrated data acquisition timing across multiple testing devices to ensure precise sensor telemetry collection.
- Managed project version control and code collaboration using Python, VS Code, and GitLab.

**Observatory Assistant — Cal Poly Physics Department** *May 2026 — Present*

- Operating 14-inch Meade LX600 ACF Telescope for public viewing events.
- Leading educational observing sessions for student groups, presenting key astrophysics concepts and managing night-sky targets.

## PROJECTS

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**Cal Poly Space Systems — Liquids Project**

- Development of bipropellant liquid rocket engines, test stand, and associated test procedures and safety
- Familiarization with Nitrous Oxide and Ethanol as propellants and work with high pressure systems
- Static pressure testing, hot fires, cold flows, and work on various projects including regenerative rocket engine and test stand maintenance

## LEADERSHIP EXPERIENCE

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**Cal Poly Space Systems Lead — SILLE (Student Introductory Liquids Learning Engine)**

- Guiding students through project to design and manufacture their own small liquid rocket engine to test up to 50lbs thrust
- Organization of over 40 students into teams, teaching basics of liquids rocketry, foundational understanding of isentropic equations and associated variables, machining, and safe testing practices
- Public speaking and proficiency in communication utilizing Microsoft teams, powerpoint, forms, and excel
- Mentorship with focus on member retention and transfer of knowledge

**Society of Women Engineers at Cal Poly**

- Advocating for and supporting women underrepresented in fields of engineering.
- Community outreach and educational opportunities to advance access to STEM and inspire the next generation of girls in engineering